ENEE631 Fall 2001
Lecture-17

Image Enhancement (2)

Min Wu

Electrical & Computer Engineering
Univ. of Maryland, College Park

☞ *http://www.ece.umd.edu/class/enee631/*
☞ *minwu@eng.umd.edu*

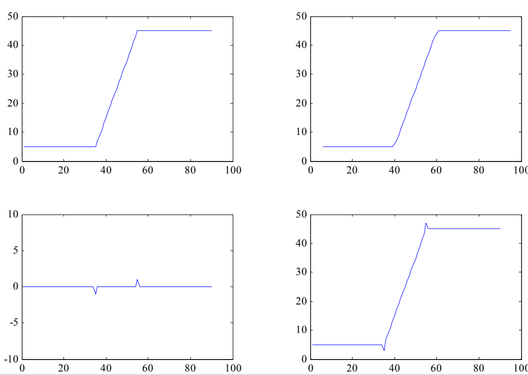
Last Time

- Pixel-based operation ~ *input-output luminance mapping*
  - Gamma correction
  - Histogram based operation
- Spatial FIR filtering using small "mask" ~ *simple to compute*
  - Spatial averaging
  - Median filtering
- Today
  - Continued on image enhancement
    - *Image sharpening*
    - *LPF/HPF/BPF*
    - *Image interpolation*
  - Edges and textures

Image Sharpening

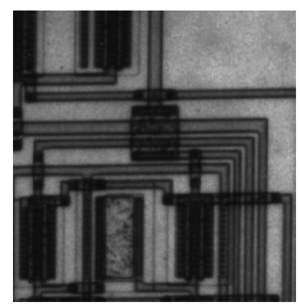
- Use LPF to generate HPF
  - Subtract a low pass filtered result from the original signal
  - HPF extracts edges and transitions
- Enhance edges

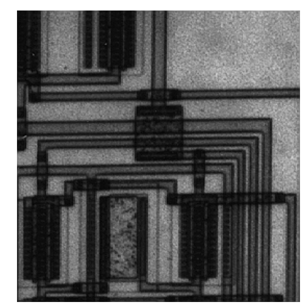
$I_0 \rightarrow I_{LP}$
$\rightarrow I_{HP} = I_0 - I_{LP}$
$\rightarrow I_1 = I_0 + a * I_{HP}$



Example of Image Sharpening

Original circuit image is from Matlab Image Toolbox.





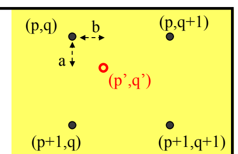
- $v(m,n) = u(m,n) + a * g(m,n)$
- Often use Laplacian operator to obtain $g(m,n)$

	-1	0	1
-1	0	-¼	0
0	-¼	1	-¼
1	0	-¼	0

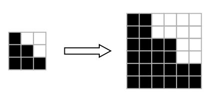
Spatial LPF, HPF, & BPF

- HPF and BPF can be constructed from LPF
- Low-pass filter
  - Useful in noise smoothing and downsampling/upsampling
- High-pass filter
  - $h_{HP}(m,n) = \delta(m,n) - h_{LP}(m,n)$
  - Useful in edge extraction and image sharpening
- Band-pass filter
  - $h_{BP}(m,n) = h_{L2}(m,n) - h_{L1}(m,n)$
  - Useful in edge enhancement
  - Also good for high-pass tasks in the presence of noise
    - *avoid amplifying high-frequency noise*
  - See also Jain's Fig. 7.25

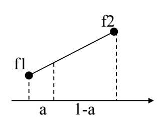
Interpolation / Zooming



- How to make up the new pixels?
- Replication according to the nearest neighbor
  - Pros: simple
  - Cons: zig-zag boundary ~ aliasing



- Bilinear interpolation
  - Extend 1-D linear interpolation
  - Do two horizontal and one vertical 1-D interpolation

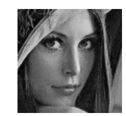


$$F(p', q') = (1-a)\,[\,(1-b)\,F(p, q) + b\,F(p, q+1)\,] + a\,[\,(1-b)\,F(p+1, q) + b\,F(p+1, q+1)\,]$$

  - For zoom in by 2 in each dimension

$$F(p', q') = 0.5 * [0.5*F(p,q) + 0.5*F(p,q+1)] + 0.5 * [0.5*F(p+1,q) + 0.5*F(p+1,q+1)]$$

Examples of Image Interpolation



4x zoom (nearest neighbor) 4x zoom (bilinear)

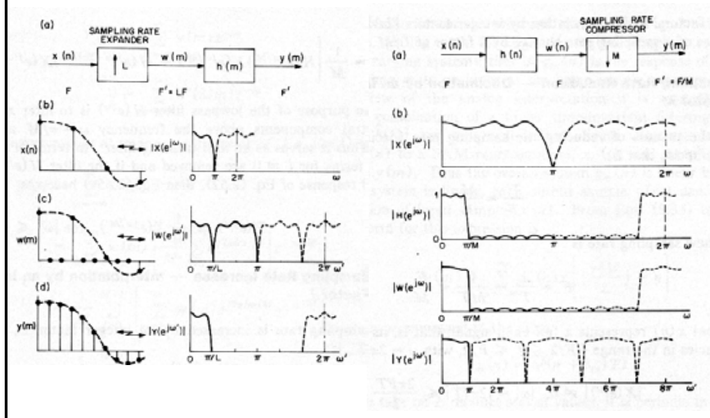
Frequency-Domain Interpretation

- Multi-rate sig. proc. (ENEE624)
- Upsampling
  - Upsampling with zero interlacing ~ replicated spectrum
  - LPF to filter out the spectra replicas in high-frequency part
  - Ideal filter vs. practical filters
    - *nearest neighbor approach for 2x zoom use*
    - *[think] what equiv. filters used for bilinear interpolation?*

1	1
1	1

- Downsampling
  - Aliasing as spectra replicas becomes closer
  - LPF to avoid aliasing
- Sampling rate conversion with rational rate M/N
  - Upsample with zero interlacing by M ➔ LPF ➔ Downsample

Frequency-Domain Interpretation (cont'd)

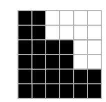


From Crochiere-Rabiner "Multirate DSP" book Fig.2.15-16

More on Interpolation

- Other filters
  - Bi-cubic interpolation (3^{rd} order polynomial)
- Interpolation that avoids blurred edges and textures
  - Edge-preserving interpolation
    - *recent research papers in ICIP*

Edges



- Edge: pixel locations of abrupt luminance change
- For binary image
  - E.g., take black pixels with immediate white neighbors as edge pixel
  - Detectable by XOR operations
- For continuous-tone image
  - Edge map ~ edge intensity + edge directions
    - *spatial luminance gradient of edge pixel is perpendicular to the edge*

Edge Detection

- Measure gradient vector
  - Along two orthogonal directions ~ usually horizontal and vertical
    - $g_x = \partial L / \partial x$
    - $g_y = \partial L / \partial y$
  - Magnitude of gradient vector
    - $g(m,n)^2 = g_x(m,n)^2 + g_y(m,n)^2$
    - $g(m,n) = |g_x(m,n)| + |g_y(m,n)|$ (preferred in hardware implement.)
  - Direction of gradient vector
    - $\tan^{-1} [\, g_y(m,n) / g_x(m,n) \,]$
- Characterizing edges in an image
  - (binary) Edge map: specify "edge point" locations with g(m,n) > thresh.
  - Edge intensity map: specify gradient magnitude at each pixel
  - Edge direction map: specify directions

Common Gradient Operators for Edge Detection

$H_1(m,n)$:

Roberts

<u>0</u>	1
-1	0

Prewitt

-1	0	1
-1	<u>0</u>	1
-1	0	1

Sobel

-1	0	1
-2	<u>0</u>	2
-1	0	1

$H_2(m,n)$:

Roberts

<u>1</u>	0
0	-1

Prewitt

-1	-1	-1
0	<u>0</u>	0
1	1	1

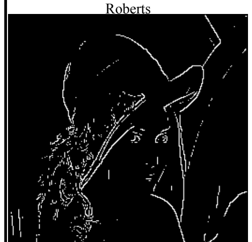
Sobel

-1	-2	-1
0	<u>0</u>	0
1	2	1

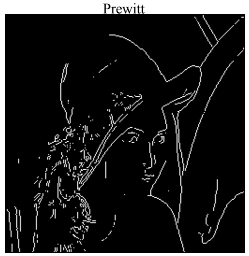
- Move the operators across the image and take the inner products
- Gradient operator is HPF in nature
  - *could amplify noise*
  - *Prewitt and Sobel operators compute horizontal and vertical differences of local sum to reduce the effect of noise*

Examples of Edge Detectors

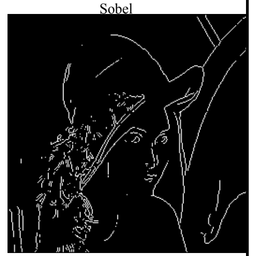
Roberts



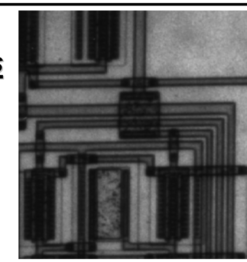
Prewitt



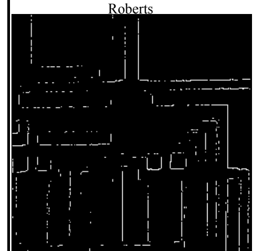
Sobel



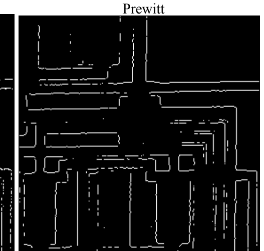
Examples of Edge Detectors



Roberts



Prewitt



Sobel

Other Edge Detection Methods

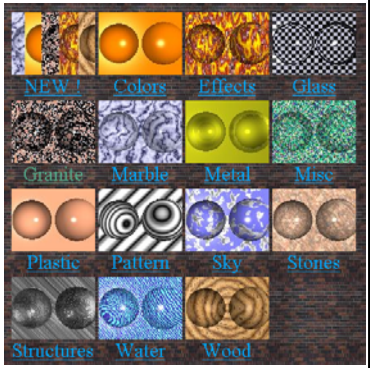
- Compass operators
  - Directly measure gradients in a selected number of directions
    - *e.g., horizontal, vertical, diagonal*
- Laplacian operators
  - [1-D case] $-d^2f / dx^2$
    - *give two extremes and one zero crossing around edges*
  - [2-D case] $\nabla^2 f = (\partial^2 f / \partial x^2) + (\partial^2 f / \partial y^2)$

0	-¼	0
-¼	1	-¼
0	-¼	0

- Improved detection in presence of noise
  - Moderate LPF before applying edge detector
- Performance comparison of edge detector ~ *Jain's Sec.9.4*

Texture

- Observed in structural patterns of objects' surfaces
  - [Natural] wood, grain, sand, grass, tree leaves, cloth
  - [Man-made] tiles, printing patterns
- "Texture" ~ repetition of "texels" (basic texture element)
  - Texels' placement could be periodic, quasi-periodic, or random



From http://texlib.povray.org/textures.html

Characterize textures

- Structural measures
  - Periodic textures
    - *Features of deterministic texel: gray levels, shape, orientation, etc.*
    - *Placement rules: period, repetition grid pattern, etc.*
  - Textures with random nature
    - *Features of texel: edge density, histogram features, etc.*
- Statistical measures ~ *mainly for textures with random nature*
  - Model as a <u>*random field*</u> ~ 2-D sequence of random variables
  - Autocorrelation function: measuring the relations among those r.v.
    - $R(m,n; m',n') = E[\, U(m,n)\, U(m',n')\,]$
    - *"wide-sense stationary":* $R(m,n;m'n') = R_U(m-m',n-n')$ *& const. mean*
  - Fit into random field models ~ *analysis and synthesis*
    - *focus on* 2^{nd}*–order statistics ~ usually two textures with same* 2^{nd}*-order prob. distribution appears very similar*

Summary

- Continued on image enhancement
  - Image sharpening
  - LPF/HPF/BPF
  - Image interpolation
- Edges
- Textures
- Next time
  - Image manipulations
- Reading Assignment
  - Jain's book 7.4, 9.4, 9.11 (2.9-2.11)